

MTH 310/520 Graph Theory: Quiz 3

Instructor: Manuj Mukherjee

TAs: Ashwin Singh & Mohammad Umar

Total: 10 points

- 1) Let $G = (V, E)$ be a graph with $\chi(G)$. Define orientations for each edge in $E(G)$, and we shall use the notation (u, v) to denote the edge $\{u, v\}$ with orientation $u \rightarrow v$. We call D to be the graph G with the oriented edges. We shall call a path $P = u_1 u_2 \dots u_m$ in G to be a directed path in D , if every edge $\{u_i, u_{i+1}\} \in E(P)$ has orientation $u_i \rightarrow u_{i+1}$. Similarly, we can define a directed cycle.

Consider the longest directed path in D which happens to contain k vertices. Let D' be the maximal spanning subgraph of D (i.e., a spanning subgraph of G with same orientations as in D) that does not contain any directed cycles, and contains this longest path. Define a function $c : V(D) \rightarrow [k]$ as follows. Fix any $v \in V(D)$, and let l be the number of vertices in the longest directed path in D' starting at v . Then, define $c(v) = l$.

a) Argue that c is well-defined, i.e., range of c is a subset of $[k]$.

b) Now, let $(u, v) \in E(D')$. Then, show that $c(u) > c(v)$.

[Hint: Let P be the longest directed path in D' starting at v . Is it possible that $u \in V(P)$?

c) Next, consider that $(u, v) \notin E(D')$. Use the maximality of D' to argue that $c(v) > c(u)$.

d) Hence, argue that c is a valid colouring of G .

e) Show that $k \geq \chi(G)$.

[1+3+4+1+1=10 points]

[CO3,CO4]